Assignment 1

" Variance and Bias (Diagram, overfit, underfit)-For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance"

Solution:

Introduction:

- Bias and Variance are two major sources of error in machine learning models.
- Bias occurs when the model makes overly simple assumptions about the data.
- High bias → Model cannot capture underlying patterns → Underfitting.
- Low bias → Model better captures patterns and reduces training error.
- Variance occurs when the model is too sensitive to small changes in the training data.
- High variance → Model fits noise along with true patterns → Overfitting.
- Low variance → Model predictions are stable across different datasets.

Bias-Variance Tradeoff:

- The total error in a **model = Bias² + Variance + Irreducible Error.**
- Increasing model complexity reduces bias but increases variance.
- Decreasing model complexity reduces variance but increases bias.
- Goal: Find the balance where both bias and variance are low for best performance.

Example:

- Predicting house prices:
  • A simple linear model → High bias, may underfit.
  • A very complex model → High variance, may overfit.
  • Optimal model → Low bias and low variance, accurately predicts prices for new houses.

Bias :

- **Bias** is the error introduced when a machine learning model makes overly simple assumptions about the data.
- **High Bias:**
  - The model cannot capture the true patterns in the data.
  - Leads to **underfitting**, where the model performs poorly on both training and test data.
  - Example: Using a straight line to fit data that follows a curve.

- **Low Bias:**
  - The model captures patterns in the data effectively.
  - Leads to better predictions if variance is also controlled.
- **Impact:**
  - High bias → High error, poor learning.
  - Low bias → Better learning and model accuracy.

Variance:

- **Variance** is the error caused when a model is too sensitive to small changes in the training data.
- **High Variance:**
  - The model fits noise along with the patterns in the training data.
  - Leads to **overfitting**, where training error is low but test error is high.
  - Example: A high-degree polynomial perfectly fitting every training point but failing on new data.
- **Low Variance:**
  - The model's predictions remain consistent even if the training data changes slightly.
  - Leads to better generalization.
- **Impact:**
  - High variance → Model does not generalize well on new data.
  - Low variance → Stable predictions and better real-world performance.

Overfitting and Underfitting:

1. **Underfitting:**
   - Occurs when a model is **too simple** to capture the patterns in the data.
   - High **Bias**, Low **Variance**.
   - Training error is high and test error is also high.
   - Example: Using a linear model to fit a curved dataset.
   - Result: Poor performance on both training and unseen data.

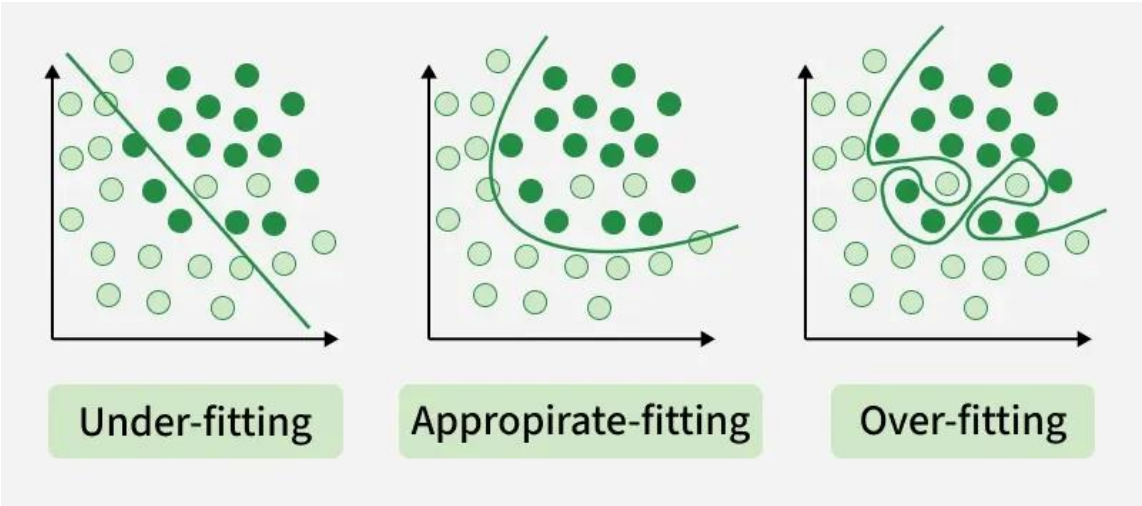
2. **Overfitting:**
    - Occurs when a model is **too complex** and fits the training data too closely, including noise.
    - Low **Bias**, High **Variance**.
    - Training error is low but test error is high.
    - Example: A high-degree polynomial that passes through all training points but fails on new data.
    - Result: Model performs well on training data but poorly on unseen data.
3. **Good Model:**
    - Balanced complexity → Low Bias + Low Variance.
    - Performs well on both training and new data.
    - Example: Properly tuned decision tree or neural network.

Model Type	Definition	Real-Life Example	Outcome
Underfitting	Model is too simple, ignores important patterns	Predicting house prices using only *size of the house*, ignoring location, age, and amenities	Poor predictions on both training and test data
Overfitting	Model is too complex, memorizes noise	A spam filter that memorizes exact words like "FREE $$$" instead of learning general spam patterns	Perfect on training emails, fails on new ones
Good Fit	Balanced complexity, captures true patterns	Netflix recommendation system that considers watch history, ratings, and similar users	Accurate suggestions that generalize well

- **Underfitting :** Straight line trying to fit a curved dataset but cannot capture the data's patterns, leading to poor performance on both training and test sets.
- **Overfitting**: A squiggly curve passing through all training points, failing to generalize performing well on training data but poorly on test data.
- **Appropriate Fitting**: Curve that follows the data trend without overcomplicating to capture the true patterns in the data.

Bias-Variance Tradeoff

- The tradeoff between bias and variance determines the total error in a model.
- Total Error = **Bias² + Variance + Irreducible Error**
- Increasing model complexity → Decreases **Bias** but increases **Variance**.
- Decreasing model complexity → Increases **Bias** but decreases **Variance**.
- Find a model with **low Bias and low Variance** to minimize total error.



- **Underfitting (High Bias):** A model that is too simple (like a straight line for curved data) misses key patterns and performs poorly on both training and testing data.
- **Overfitting (High Variance):** A model that is too complex (like a high-degree polynomial) learns noise, fits training data too closely, and performs poorly on new data.
- **Ideal Fit (Balanced):** A moderately complex model captures the main trend without following noise, giving good performance on both training and testing data.

Importance:

a. Helps select the right model complexity.
b. Ensures the model generalizes well to unseen data.
c. Guides tuning of parameters like tree depth, polynomial degree, or number of neurons.

Best Fit Model (Low Bias, Low Variance):

- A Best Fit model is one that **captures the true patterns in the data** without being too sensitive to noise.
- Achieved by having **low Bias** and **low Variance**.
- Accurately represents the relationship in the data.
- **Performs well on both training data and unseen test data**.
- Avoids **underfitting** (high bias) and **overfitting** (high variance).

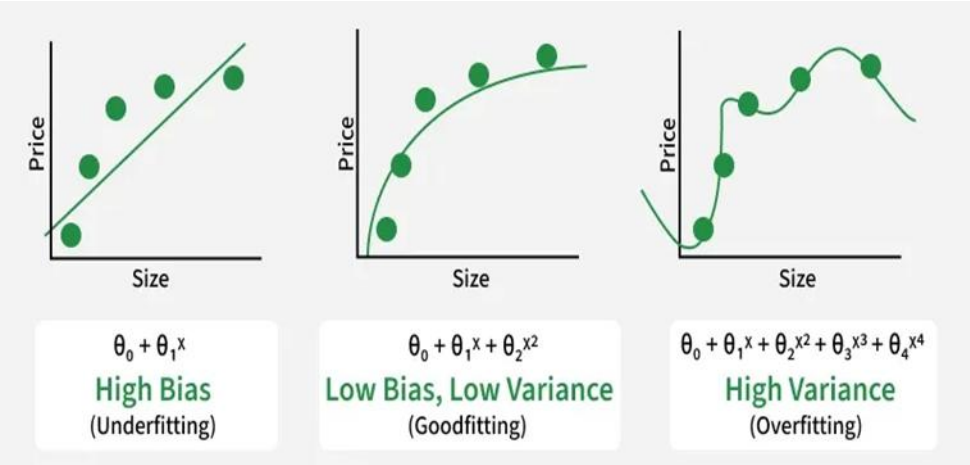
- Choose an appropriate model complexity (not too simple or too complex).
- Use techniques like cross-validation to tune parameters.
- Remove irrelevant features and select meaningful ones.
- Apply regularization methods (e.g., L1, L2) to reduce overfitting.

Example:

- Predicting house prices:
  • Low Bias → Captures key features like location, size, and number of rooms.
  • Low Variance → Performs well on new houses, not just the training data.

Low Bias + Low Variance = Best Fit Model → Optimal performance, good generalization, minimal total error.

Diagram – Bias vs Variance (Overfit, Underfit, Best Fit):



Explanation of Bias-Variance Diagram:

High Bias (Underfitting)

- The straight line cannot capture the true pattern of the data.
- The model is **too simple** and ignores important trends.
- Both **training and test errors are high**.
- Example: Predicting house prices using only size while ignoring location or other factors.

Low Bias, Low Variance (Best Fit / Good Fit):

- The curve fits the data well without capturing noise.
- The model captures the **true underlying pattern**.
- Both **training and test errors are low**.
- This is the **optimal model** that generalizes well to new data.

High Variance (Overfitting):

- The curve fits every single training data point, including noise.
- The model is **too complex** → performs very well on training data but poorly on unseen data.
- Training error is low, but test error is high.
- Example: A high-degree polynomial that fits all past house prices but fails on new houses.

Conclusion:

In machine learning, a model's performance depends on its ability to make accurate predictions on new data. Two important factors that affect this are **bias** and **variance**. Bias measures the error due to overly simplistic assumptions in the model. High bias means the model is too simple and cannot capture the underlying patterns, leading to **underfitting**. Variance measures how much the model's predictions change when trained on different datasets. High variance means the model is too sensitive to the training data, leading to **overfitting**.

The key is to find a **balance between bias and variance**, known as the **bias-variance tradeoff**, so that the model generalizes well. A **best fit model** has **low bias and low variance**, accurately capturing patterns while being stable across datasets. Techniques like cross-validation, regularization, and careful selection of model complexity help achieve this balance.

1. **Bias:** Error due to oversimplified assumptions → leads to underfitting.
2. **Variance:** Error due to sensitivity to training data → leads to overfitting.
3. **High bias → Underfitting**, high variance → Overfitting.
4. **Best fit model:** Low bias + Low variance.
5. **Goal:** Accurate predictions on new, unseen data.
6. **Solution:** Balance bias and variance using cross-validation, regularization, and proper model choice.